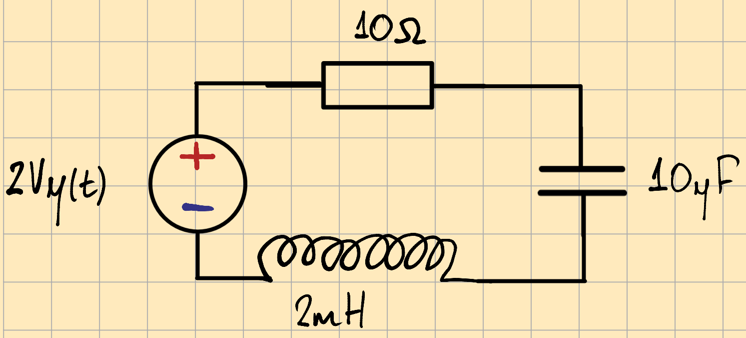
Apunte #7 RLC

¿Como se analiza un cto de Segundo Orden?

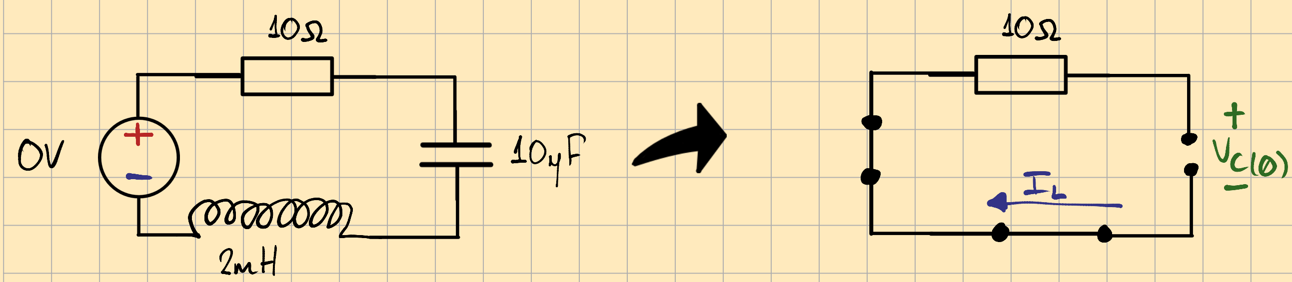


Evaluamos el cto en el momento antes del cambio ($t=\emptyset$)

$t=\emptyset^-$

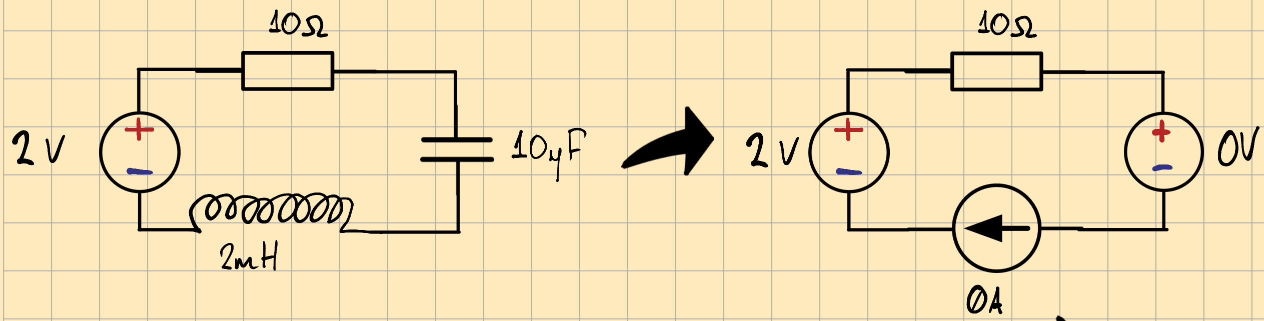
$V_{C(\emptyset)}=\emptyset V$

$I_{L(\emptyset)}=\emptyset A$

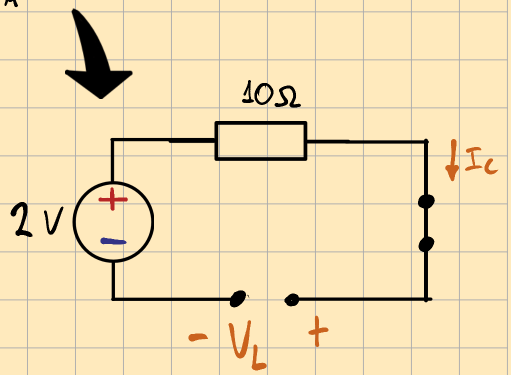


Evaluamos el cto en el momento despues del cambio ($t=\emptyset$)

$t=\emptyset^+$



	Relación		
$I_C=0A$	$I_{C(t)}=C\dfrac{dV_{C(t)}}{dt}$	$\dfrac{dV_{C(t)}}{dt}=\dfrac{I_{C(t)}}{C}$	$\dfrac{dV_{C(0^-)}}{dt}=0V/s$
$V_L=2V$	$V_{L(t)}=L\dfrac{dI_{L(t)}}{dt}$	$\dfrac{dI_{L(t)}}{dt}=\dfrac{V_{L(t)}}{L}$	$\dfrac{dI_{L(0^-)}}{dt}=1000A/s$



Evaluamos el cto mucho tiempo despues ($t=\infty^+$) (Rta Forzada)

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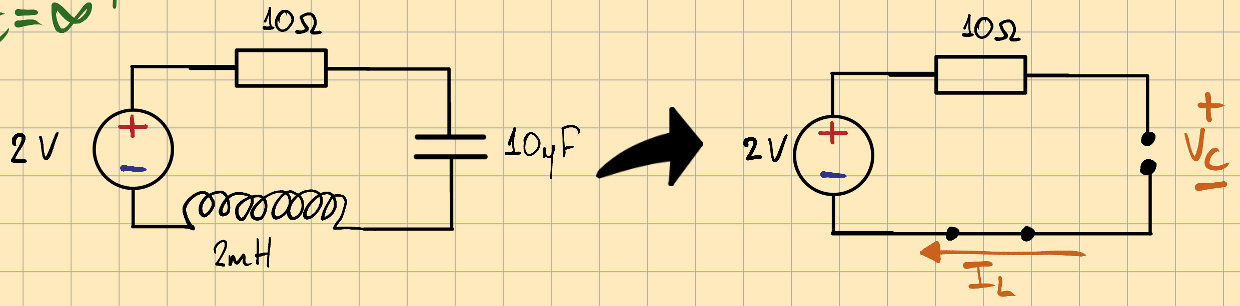
$t=\infty^+$

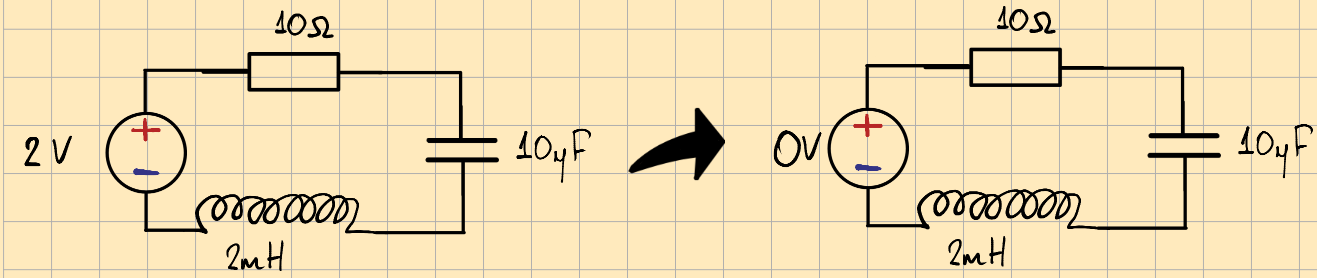


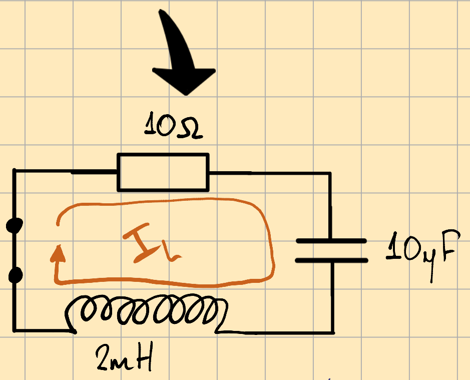
$V_c = 2V$

$I_L = 0A$

Ahora analizamos la rta natural

La Rta Natural es tiempo positivo y con todas las fuentes apagadas





LVK:

$$I_{L(t)} \cdot R + V_{C(t)} + V_{L(t)} = 0$$

$$I_{L(t)} \cdot R + V_{C(t)} + V_{L(t)} = 0 \qquad V_{L(t)} = L\frac{dI_{L(t)}}{dt}$$

$$I_{L(t)} \cdot R + V_{C(t)} + L\frac{dI_{L(t)}}{dt} = 0$$

$$I_{(t)} = C\frac{dV_{C(t)}}{dt}$$

$$\int I_{(t)}\,dt = C \cdot \int dV_{C(t)}$$

$$\frac{1}{C}\int I_{(t)}\,dt = \int dV_{C(t)}$$

$$\frac{1}{C} \cdot \int I_{(t)}\,dt = V_{C(t)} + C$$

$$\boxed{\frac{1}{C} \cdot \int I_{C(t)}\,dt + C = V_{C(t)}}$$

$$I_{L(t)} \cdot R + \frac{1}{C}\int I_{L(t)}\,dt + C + L\frac{dI_{L(t)}}{dt} = 0$$

$$\frac{d\left(I_{L(t)} \cdot R + \frac{1}{C}\int I_{L(t)}\,dt + C + L\frac{dI_{L(t)}}{dt} = 0\right)}{dt}$$

$$\frac{dI_{L(t)}}{dt} \cdot R + \frac{1}{C}\,\frac{d\int I_{L(t)} \cdot dt}{dt} + \frac{d \cdot C}{dt} + L \cdot \frac{d^2 I_{L(t)}}{dt^2} = \frac{d\,0}{dt}$$

$$\frac{dI_{L(t)}}{dt} \cdot R + \frac{1}{C}\,I_{L(t)} + 0 + L\frac{d^2 I_{L(t)}}{dt} = 0$$

$$\frac{dI_{L(t)}}{dt} \cdot R + \frac{1}{C}\,I_{L(t)} + L\frac{d^2 I_{L(t)}}{dt} = 0$$

$$L\frac{d^2 I_{L(t)}}{dt^2} + R\frac{dI_{L(t)}}{dt} + \frac{1}{C} \cdot I_{L(t)} = 0 \qquad S^2 = \frac{d^2 I_{L(t)}}{dt^2}\,,\; S^1 = \frac{dI_{L(t)}}{dt}\,;\; S^0 = 1 = I_{L(t)}$$

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$$L\cdot s^2 + R\cdot s + \frac{1}{C} = 0$$

$$s^2 + \frac{R}{L}s + \frac{1}{LC} = 0 \qquad s^2 + 2\alpha s + \omega_0^2 = 0$$

Hallamos α y ω_0

$$2\alpha = \frac{R}{L} \qquad \omega_0^2 = \frac{1}{LC} \qquad \alpha = 2{,}5K$$

$$\alpha = \frac{R}{2L} \qquad \omega_0 = \sqrt{\frac{1}{LC}} = \frac{1}{\sqrt{LC}} \qquad \omega_0 = 7{,}07K$$

Determinar el caso

Caso	Condición	Raíces	$x_n(t)$	
Sobreamortiguado	$\alpha > \omega_0$	s_1, s_2 reales y distintas	$A_1e^{s_1t} + A_2e^{s_2t}$	$s_{1,2} = -\alpha \pm \sqrt{\alpha^2 - \omega_0^2}$
Críticamente amortiguado	$\alpha = \omega_0$	$s = -\alpha$ (doble)	$(A_1 + A_2t)e^{-\alpha t}$	
Subamortiguado	$\alpha < \omega_0$	$s = -\alpha \pm j\omega_d$	$e^{-\alpha t}(A_1 \cos \omega_d t + A_2 \sin \omega_d t)$	$\omega_d = \sqrt{\omega_0^2 - \alpha^2}$

$$\alpha < \omega_0 \qquad \omega_d = 6{,}6K$$

$$\left. I_{L(t)} = e^{-\alpha t}\left(A_1\cos(\omega_d \cdot t) + A_2\sin(\omega_d \cdot t)\right) \right\} \text{Rta. Nat}$$

Rta completa = Rta Nat. + Rta. Forzada

$$I_{L(t)} = e^{-\alpha t}\left(A_1\cos(\omega_d \cdot t) + A_2\sin(\omega_d \cdot t)\right) + 0A$$

derivo $I_{L(t)}$ con respecto a t

$$\frac{dI_{L(t)}}{dt} = -\alpha e^{-\alpha t}\left(A_1\cos(\omega_d \cdot t) + A_2\cdot\sin(\omega_d t)\right) + e^{-\alpha t}\left[-A_1\omega_d\sin(\omega_d \cdot t) + A_2\omega_d\cos(\omega_d t)\right]$$

Evaluo en $t=0$

$$0A = e^{-\alpha 0}\left(A_1\cos(\omega_d \cdot 0) + A_2\sin(\omega_d \cdot 0)\right) + 0A$$

$$0 = 1\left(A_1 \cdot 1 + A_2 \cdot 0\right) = A_1$$

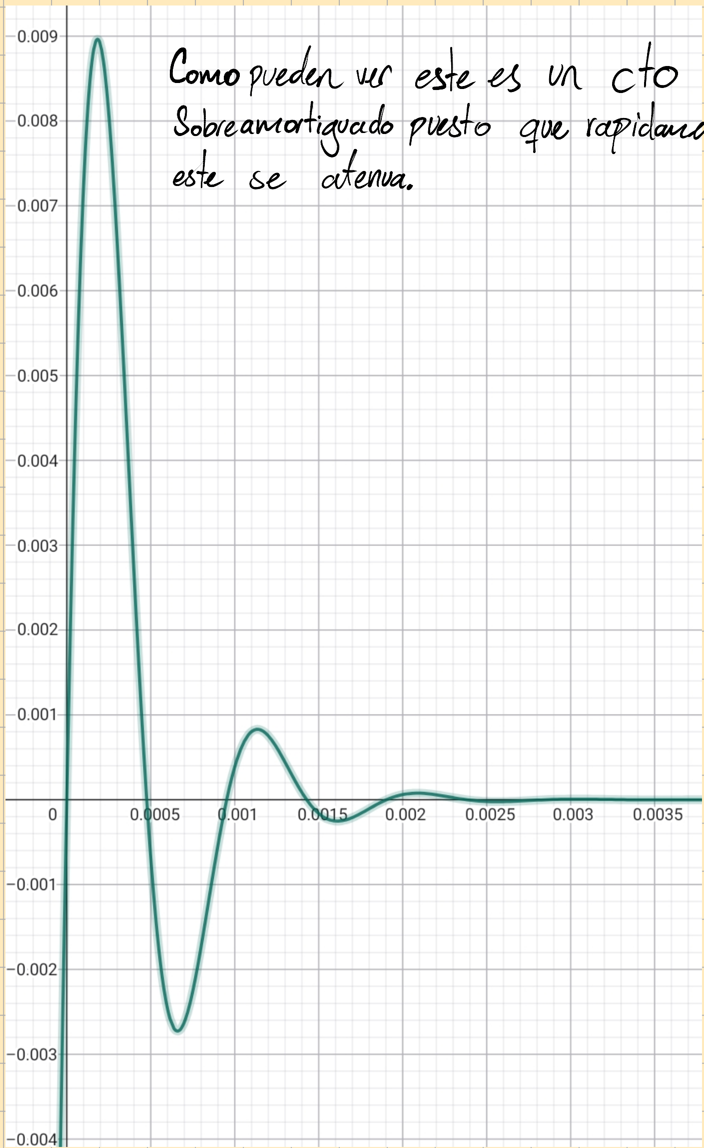
$$100A/s = -\alpha e^{-\alpha 0}\left(\overset{0}{A_1}\cos(\omega_d \cdot 0) + A_2\cdot\sin(\omega_d \cdot 0)\right) + e^{-\alpha 0}\left[-\overset{0}{A_1}\omega_d\sin(0 \cdot t) + A_2\omega_d\cos(\omega_d 0)\right]$$

$$100A/s = -\alpha\cdot 1\cdot(0+0) + 1\cdot(0 + A_2\cdot\omega_d\cdot 1) = A_2\cdot\omega_d$$

$$\frac{100A/s}{\omega_d} = A_2 = 15{,}15m$$

$$I_L(t) = e^{-2,5K\cdot t}\left(0\cdot\cos(6,6K\cdot t) + 15,15m\ \sin(6,6K\cdot t)\right)$$

$$I_L(t) = 15,15m\cdot e^{-2,5K\cdot t}\cdot\sin(6,6K\cdot t)$$



Como pueden ver este es un cto Sobreamortiguado puesto que rapidamente este se atenua.

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